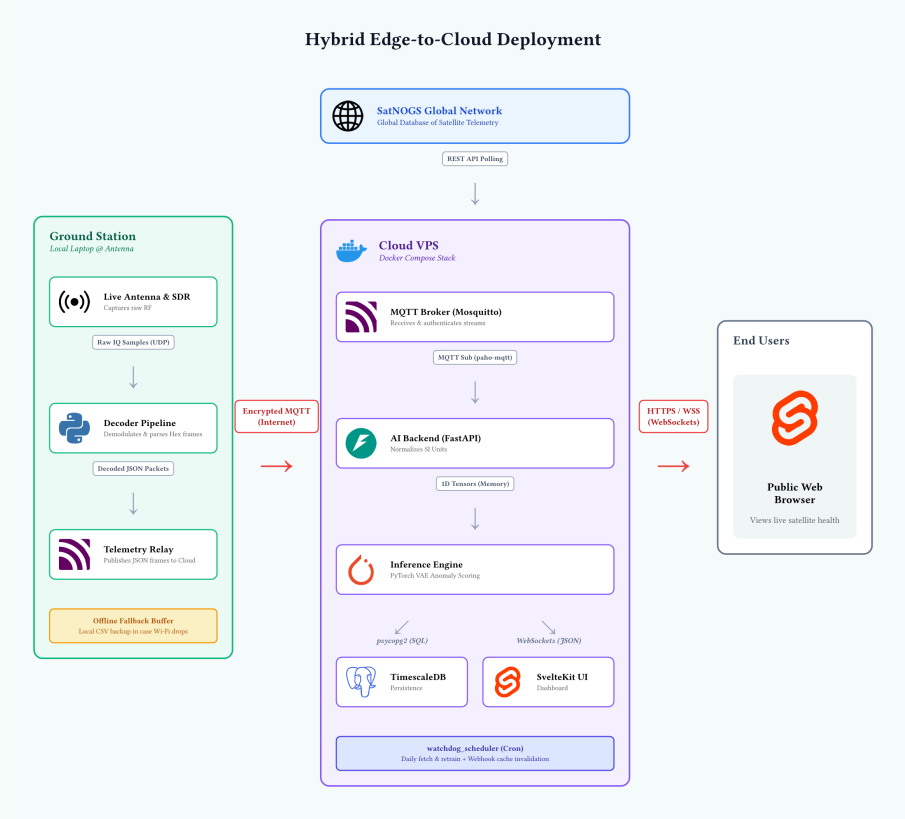


UWE-4 System Architecture

Docker Topologies & Edge Integration

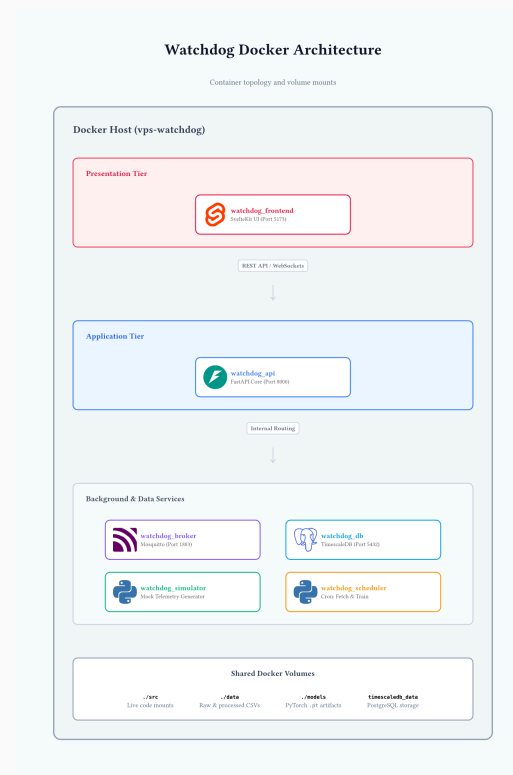
Architecture & Integration

The Edge-to-Cloud Topology



The Dockerized Application Stack

Our Cloud VPS is fully containerized with an automated nightly training job.



Implementation Logic

The Shared Core Pipeline

We use the “Shared Core” (`src/gr_sat/telemetry.py`) which acts as the universal adapter between raw bits and our AI model.

Pipeline Steps:

1. **Ingest:** Read raw hex frames (SatNOGS/SDR).
2. **Parse:** Kaitai Structs via `satnogs-decoders`.
3. **Normalize:** Convert binary fields to SI Units (V, A, °C).
4. **Validate:** Check against physical limits.

Current Status:

- Core Logic: **Operational (Shared Core)**
- Decoder Ecosystem: **satnogs-decoders**
- Coverage: **1 production decoder in-repo (UWE-4)**
- Online Runtime: **Implemented (FastAPI + TimescaleDB + SvelteKit UI)**

Because real anomaly data is rare, we validate the model using **Synthetic Fault Injection** on clean data.

Accuracy (Recall / FPR)

1. **Panel Failure:** High temp, but negative current.
2. **Thermal Runaway:** Inject large positive battery-temp steps.
3. **Sensor Stuck:** Historical notebook experiment, not in current shipped benchmark.

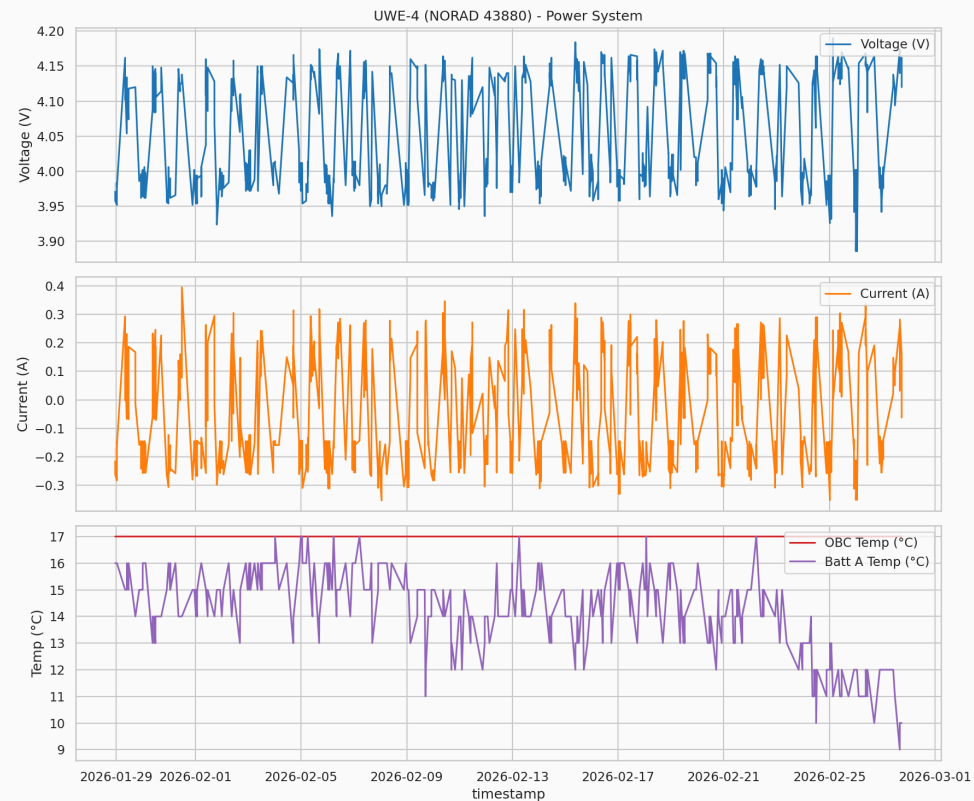
Edge Targets (Online Runtime)

Integrated with Dockerized Backend / MQTT Broker.

- **Latency:** target < 10 ms per frame.
- **Memory:** target < 5 MB model footprint.

Data Verification (UWE-4)

Telemetry extraction from UWE-4 (NORAD 43880). Perfect mapping from `satnogs-decoders` to our Golden Features (Volts, Amps, °C).



The Inspector Tool

We built an interactive debugger (telemetry_inspector) to verify decoders against real historical data.

The screenshot displays the 'Telemetry Inspector' tool interface, which is part of the 'WATCHDOG' system. The interface is divided into several sections:

- Left Sidebar:** Contains navigation links for 'LIVE DASHBOARDS' (Dashboard Home, Operations, Live Watcher, Inspector, Analytics, ML Interface), 'NOTEBOOKS' (EDA Report, Model Analysis), and a 'Light Mode' toggle.
- Header:** Labeled 'DEEP DIVE' and 'Telemetry Inspector'.
- Filters:** Includes a 'SATELLITE FILTER' set to 'All Satellites' and a 'FETCH LIMIT' set to '50 frames', with a 'Fetch Data' button.
- RECENT PACKETS:** A list of recent packets from satellite 'NORAD-43880', each marked as 'Complete'.
- FRAME DETAILS:** Contains three main sections:
 - RAW PAYLOAD (HEX):** Displays a hexadecimal string: 88 88 60 AA AE 8A 60 88 A0 60 AA AE 90 E1 03 F0 25 82 7B 87 02 01 01 00 0E 2E 1F 00 05 AF 23 AC D4 49 F4 9E 01 00 F0 00 00 03 1F D5 29 04 00 0A 7F 0B 54 0A 59 25 00 04 10 A6 00 3C 10 D6 04 11 00 13 14 00 14 15 10 8B 9B D0.
 - RAW KAITAI DECODED STRUCT:** A table of decoded fields:

dest_callsign	"D00LME"
src_callsign	"D00LMEH"
src_ssld	0
dest_ssld	0
ctl	3
pid	240
beacon_header_flags1	37
beacon_header_flags2	130
beacon_header_packet_id	34683
beacon_header_frame_id	0
 - NORMALIZED GOLDEN FEATURES:** A grid of feature values:

BATT_VOLTAGE: 4.1280	BATT_CURRENT: 0.2630
BATT_A_VOLTAGE: 4.1000	BATT_B_VOLTAGE: 4.1560
BATT_A_CURRENT: 0.0370	BATT_B_CURRENT: 0.1660
POWER_CONSUMPTION: 1.2380	TEMP_OBC: 17.0000
TEMP_BATT_A: 11.0000	TEMP_BATT_B: 10.0000
TEMP_PANEL_Z: 20.0000	UPTIME: 272853.0000

Summary of Phase 3

1. **Targeting:** Locked onto **UWE-4** (43880) as our primary target.
2. **Pipeline:** Processing is operational, successfully generating structurally clean ML inputs.
3. **Data Strategy:** Fetching 180+ days of data to capture seasonal variations.